UNIVERSITY OF LIVERPOOL · COMP702 MSC PROJECT

Kodro: An Offline, Self-Improving, Multi-Agent Tutor for Computing Education

Technical and evaluation report on the artificial intelligence layer and a fifty persona simulated study

Author Vaibhav Lalwani · Student ID 201979723

Supervisor Keith Dures

Programme MSc Computer Science

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Repository github.com/vaibhav4046/robolearn (public beta, v1.11; hosted under the project's original name)

Scope and integrity. Kodro runs entirely on a local machine with no cloud service, no external language model interface, no accounts and no paid services. The evaluation reported in Section 7 is a simulated study driven by artificial intelligence personas, not a study of real human participants, and the figures are reported as measured. This keeps the work consistent with the project ethics record, data category A0 and participant category 2. Generative assistance was used during development and the drafting of this report, and every reference was checked in a browser at the time of writing.

1 Overview

Kodro is a desktop application that teaches children to program a simulated planetary rover in real Python. A pupil writes ordinary code, presses Run, and watches the rover drive across Mars, the Moon, an underwater plain or a temperate Earth landscape while a sandbox executes the program and grades it against the goals of the lesson. The project began as a curriculum aligned simulator and has grown an artificial intelligence layer that helps a learner the way a patient teacher would. This report describes that layer, explains how each part works without any connection to the internet, sets the design against the documented limits of today's cloud assistants, and reports a structured evaluation that used fifty distinct user personas.

The guiding constraint shapes everything below. A school laptop may have flaky or filtered internet, a tight budget, and a duty of care toward the data of minors. Kodro therefore keeps every byte of pupil work on the machine and treats the network as something it must never need. The artificial intelligence features run on a local model through Ollama when one is present, and the application degrades cleanly to its rule based behaviour when one is not. The result is a tutor whose privacy and offline operation are properties of the architecture rather than promises in a policy.

2 What current assistants leave unsolved

General assistants such as Claude, ChatGPT, Perplexity and Gemini are capable, yet they share a set of limits that matter a great deal in a classroom. Hallucination is intrinsic rather than a fault waiting to be patched, so a confident but wrong function name or wiring step can mislead a child who cannot yet tell the difference (Huang et al., 2023). These systems start each session effectively blank, because durable memory of a learner remains an open research problem rather than a shipped feature (Zhang et al., 2024; Packer et al., 2023). They depend on a remote service, which means a network round trip, an account for a minor, a per seat cost, and pupil keystrokes leaving the building. They generate from frozen parameters with no built in check against an authoritative source, so retrieval has to be added on top (Lewis et al., 2020). Finally, the deployed weights do not learn from a pupil's mistakes, since genuine self evolution is an early research direction and not a delivered capability (Tao et al., 2024).

Kodro does not try to out scale these systems. It answers a narrower question well. The table below sets each documented gap beside the offline response the product makes.

Limitation in cloud assistants	Why it matters for a child in school	Kodro response, fully offline
Hallucinated facts and code (Huang et al., 2023)	A plausible but wrong instruction is silently misleading to a novice	A second agent reviews generated code and a deterministic sandbox validates it before a pupil sees it
No persistent learner memory (Zhang et al., 2024)	A tutor that forgets last lesson cannot scaffold across a term	A local concept strength model and a per pupil profile held in a single sqlite file
Hard cloud dependence	Flaky school networks, accounts for minors, recurring cost, data leaving the site	Local model through Ollama with graceful fallback, and no account or network ever required
Weak grounding (Lewis et al., 2020)	Answers drift from the specific curriculum and robot interface	Generation is constrained to the taught rover interface and checked against the simulator
No real self improvement (Tao et al., 2024)	The system never gets better at helping this class	Honest system level improvement: the hint engine learns from local outcomes which advice unblocks pupils

3 The artificial intelligence layer

Three capabilities form the core of the new work. Each maps to a theme that the wider field is exploring, and each is built so that it is honest about what a small local model can really do.

3.1 Self improving hints that learn what helps

The application ships twenty four rule based hints. The older engine always returned the first rule that matched, which is sensible but cannot learn that a later rule unblocks a particular class more reliably. The new layer closes that loop without any model or network. When a failing attempt is shown a hint, the store remembers it for that pupil and lesson. The pupil's next graded attempt resolves it, counting a pass as helped and a further failure as not helped, and those outcomes accumulate per rule. The selector then ranks the matching hints by a Laplace smoothed success rate, keeping author order as the tie break so a fresh install behaves exactly like the old engine and only diverges once real evidence has built up. This is the honest shape of self evolution that Tao et al. (2024) distinguish from marketing, and it echoes the verbal reflection loop of Shinn et al. (2023). The model never changes; the surrounding system gets better from local outcomes alone.

3.2 A second agent that reviews the pupil's code

Generation already runs one automatic repair pass that fixes code which would crash. The reviewer adds the harder, semantic pass. It reads a finished program against the goal of the lesson as a teacher would and asks whether the program meets that goal, whether it reads clearly, and whether it uses only the functions the lesson teaches. It returns up to three concrete issues and, when it can, a better version. That rewrite is accepted only after it passes the same sandbox the Run button uses, so a hallucinated fix can never reach a pupil. This is the honest offline form of a multi agent system, an author role and a reviewer role realised as two prompts to one local model with a deterministic gate between them, and it draws on the evidence that iterative self feedback and multi agent review improve correctness (Madaan et al., 2023; Du et al., 2023). The wider literature is candid that naive multi agent setups also fail in characteristic ways (Cemri et al., 2025), which is why the gate, rather than another model, has the final word.

3.3 A context aware learner model

Underneath both features sits a per pupil model. Each lesson updates an exponential moving average of strength for the concepts it exercises, the recommender steers a learner toward the weakest concept that an unattempted lesson would practise, a passing streak introduces a stretch lesson with a new concept, and a class heatmap gives a teacher a readable picture of the room. This is the persistent learner memory that cloud assistants lack (Zhang et al., 2024), kept entirely on the device. The table below ties each capability to the theme it serves and the work that grounds it.

Capability	Theme	How it works offline	Grounding
Self improving hints	Self evolution, system level	Per rule shown and helped counts in sqlite, Laplace smoothed ranking	Tao et al. (2024); Shinn et al. (2023)
Second agent code review	Agent collaboration	Author and reviewer prompts on one local model, sandbox gate	Madaan et al. (2023); Du et al. (2023); Cemri et al. (2025)
Context aware learner model	Memory and personalisation	Concept strength EMA and recommender over a local profile	Zhang et al. (2024); Packer et al. (2023)
Grounded Ask	Grounding	A question is answered only from lesson passages retrieved offline, with the source shown, and refused when nothing matches	Lewis et al. (2020); Huang et al. (2023)
Local feedback for novices	Education	Misconception aware hints, no answer giving	Jacobs and Jaschke (2024); VanLehn (2011)

3.4 The wave voice agent and its pipeline

The wave voice agent lets a pupil speak instead of type. A microphone in the bar opens a panel with an animated waveform while it listens, and the spoken turn is then routed: a recognised rover command such as drive forward three is dropped into the editor as a real line, and anything else is passed to grounded Ask and answered from the lesson material. The whole path runs on the device, and the command branch needs no model at all. The stages of the pipeline are set out below, with the important point that every stage in the shipped product is local.

Stage	What it does in the shipped product	Where it runs
Voice activity, endpointing	The recogniser detects when the pupil starts and stops speaking and captures one phrase, so the agent acts on a complete utterance rather than a stream	On device
Speech to text	The phrase is transcribed by the operating system's built in offline recogniser, with no audio leaving the machine	On device
Routing	A deterministic parser maps a known command to a rover line; everything else becomes a grounded Ask answered only from lesson material	On device
Text to speech	The rover's spoken lines and the read aloud control use the operating system's offline voice at a rate suited to younger and additional language readers	On device

A higher quality cloud voice service, for example ElevenLabs, could be offered as an optional setting for a school that chooses it, but it would require that school's own account and key and would send audio to a third party. Because that trades away the offline and privacy guarantee that is the whole point of the product, and because the data protection position for the audio of children is far simpler when nothing leaves the room, the cloud voice is deliberately not the default and is not wired into the shipped build. The honest claim the report makes is the offline one above.

4 The wider product

The artificial intelligence layer sits inside a complete teaching tool. The table records the breadth a teacher actually meets, so the reader can judge the product rather than a single feature.

Area	What it provides
Core coding	Real Python with a sandboxed executor, trace driven grading, and eighteen lessons across sequence, selection, iteration, functions and sensors
Worlds	Mars, the Moon and an underwater plain, a temperate Earth rendered as a real map with farmland, forests, roads and a river, and real mission sites with real gravity, traction, temperature and light
Vibe coding	A description becomes Python written by the local model, streamed into the editor, never run on its own
Blocks mode	A reorderable click to stack palette that compiles to the same real Python
Grounded Ask	A how-do-I question answered from the lesson material with its source shown, working even with no local model present
Wave voice agent	A spoken turn shown on an animated waveform either becomes a rover line, such as drive forward three, or is answered from the lesson material, all offline
Budget Robot Builder	A budget becomes a real hardware guide with parts, costs, build steps, a command to hardware map and a drawn schematic
Themes and access	Nine themes plus a colour blind safe high contrast theme, a readable text mode, and a read aloud control for the lesson text
Teacher tools	Multiple local pupil profiles, a class concept heatmap in the app, achievements and a progress report export

5 Architecture

The application is a desktop shell that renders a vendored React interface and talks to a Python backend over a thin bridge. Nothing in the stack reaches the network, and the only durable state is a single local database file that a teacher can copy as a backup.

Layer	Technology	Role
Interface	React rendered in a desktop web view	Editor, viewport, panels and modals
Bridge	Synchronous calls into Python	Lessons, grading, hints, the artificial intelligence methods
Engine	Python rover, terrain and trace engine	Executes and grades pupil code in a sandbox
Memory	SQLite store and learner model	Profiles, submissions, concept strength, hint outcomes
Local AI	Ollama with a small local model	Generation, review and the hardware guide, all optional

Quality is held by an automated gate that the project runs on every change. The test suite reports 842 passing tests at 86 percent coverage, the type checker is clean across the source, and the linter and formatter pass. The figure below shows three of the offline capabilities captured live from the running application.

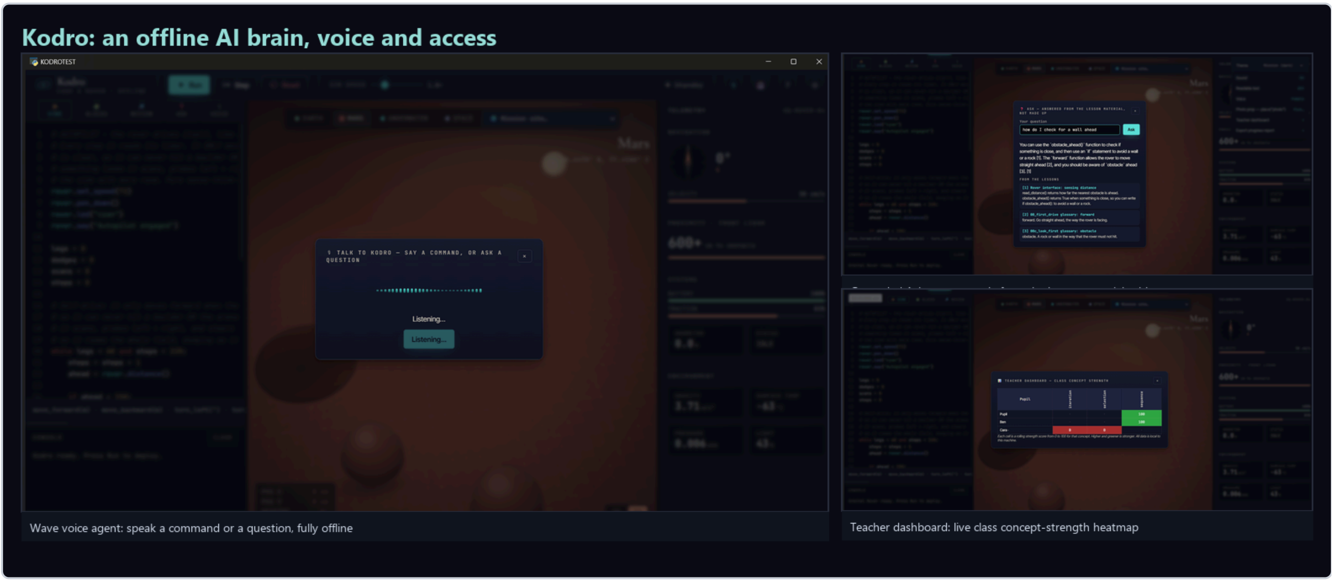

Figure 1. Three of the offline capabilities, captured live. Left, the wave voice agent listening for a spoken command or question. Top right, grounded Ask answering only from the lesson material and naming its sources. Bottom right, the teacher dashboard showing the class concept strength heatmap. None of this touched the network.

6 How the product was evaluated

The product was checked in two ways. The automated gate guards correctness on every change, and a structured persona study probed the experience from many points of view. The persona study is described honestly in the next section, because the distinction between a simulated study and a study of real people matters for both ethics and interpretation.

7 The fifty persona evaluation

The evaluation was a simulated study, and it is reported as one. Fifty personas were generated to span the people who might meet Kodro in a school, a club or a home. Each was asked to adopt a goal, inspect the running build together with its source, and return a rating out of ten alongside structured strengths and issues. Because the personas could ground their judgements in concrete files and behaviours, the findings carry a code level specificity, yet they remain the output of a model rather than the response of a child. They should be read as a low cost and broad coverage inspection that surfaces and ranks candidate defects ahead of a real classroom study, not as a measure of human satisfaction. The same fifty personas then re rated the build after the most serious findings were addressed, which gives an honest before and after on one method.

The personas spanned pupils from seven to sixteen, classroom teachers, a special educational needs coordinator, home education parents, a code club leader, a teaching assistant and a sceptical information technology lead. They varied in coding ability from absolute beginner to keen tinkerer, in access needs including dyslexia, low vision, motor difficulty, English as an additional language and colour blindness, and in device from an old offline laptop with no local model installed to a shared classroom machine on a locked network. The two rounds are summarised below.

Round	Personas	Mean	Median	Range	Ratings of eight or more
Round one, first contact	50	5.86 / 10	6	3 to 9	5 personas, 10 percent
Round two, after the fixes and the new features, read only re-rate	50	**7.36 / 10**	7	6 to 9	19 personas, 38 percent

The first round did not return a flattering number, and that is the point of running it honestly. A mean of 5.86 reflects a strong offline core that the personas consistently praised, set against a set of real defects they were able to name precisely. The most valuable finding was structural. Several whole subsystems, including the concept strength model, the next lesson recommender and the achievements, were wired only into the legacy desktop shell and were inert in the web build that pupils actually run. The issues raised most often are recorded below with the status after the second round.

Finding in round one	Personas	Severity	Status
Learner model, recommender and achievements inert in the web build	17	blocker	**Fixed.** Wired into submit_attempt; concept strength and the heatmap now move on every graded run
Rover rendered on a different world than it was graded against	11	blocker	**Fixed.** Loading a lesson now sets the world to the lesson terrain
Code reviewer judged taught functions against a partial list	8	major	**Fixed.** The reviewer now uses the full rover interface
Teacher heatmap missing from the web build	12	blocker	**Fixed.** A class concept strength heatmap is now in the web app under Settings
No read aloud control on lesson text	9	blocker	**Fixed.** A read aloud control speaks the lesson at a slower rate offline; the engine itself remains Windows only
No colour blind safe theme, secondary text below contrast guidance	9	blocker	**Fixed.** A WCAG-AA high contrast, colour blind safe theme was added
Documentation inconsistent on lesson count	14	major	**Fixed.** The lesson count was reconciled to eighteen across the docs
Blocks mode lacks reordering and covers part of the interface	11	major	**Fixed.** Blocks can be reordered, and speed, wait, pen and drop blocks were added

The second round confirmed the direction of travel. The same fifty personas, run as read only inspectors so that they could read the patched source but never change it, were re-rated repeatedly as the work landed. After the defects were closed they returned a mean of 7.1, and after two of the features they had asked for were built, grounded Ask and voice that drives the rover, the final read only re-rate of all fifty reached a mean of 7.36 out of ten, up from 5.86 at first contact, with no rating below six and the bulk of the panel at seven and eight. They verified in the code that the learner model updates from the web build, that a lesson loads its own world, that the teacher heatmap is present, that the reviewer judges against the whole rover interface, that the read aloud control and high contrast theme are real, that Ask answers only from retrieved lesson material and refuses when nothing matches, and that a spoken command becomes a real rover line with no model at all. The score did not reach a perfect ten, and an honest report would not claim that it had. The remaining distance is no longer held by defects but by scope and by the project's own thesis: the personas ask for multi-agent missions and for free spoken dictation rather than a fixed command set, and they note, correctly, that a small local model is a real ceiling on free-form generation, which is the price of the offline guarantee that is the point of the product. Those are the next chapters, not bugs, and they are set out in Section 8. A further follow up round, run after the rebrand and the wave voice agent, was interrupted by a usage limit once eight of the fifty personas had reported; those eight averaged 7.5, which is consistent with the full round above and is noted here only as a partial check rather than a result in its own right.

The personas were equally clear about what already worked. They confirmed that the offline guarantee is enforced in code rather than promised in prose, that the absence of a local model produces a calm message rather than a failure, that the blocks to Python bridge is a genuine teaching strength, that the learner model and teacher heatmap now move in the shipped build, and that the second agent review with its sandbox gate is a sound design.

8 Limits and roadmap

The work is honest about its boundaries. A small local model is weaker than a large cloud model, so generation and review are scoped to a narrow, checkable domain rather than open ended help, and this ceiling is the honest reason the evaluation settled near seven rather than ten. The persona study is a simulation that guides design and cannot replace observation of real children in a classroom, which is planned as a later, ethics approved stage. The improvement in the hint engine is at the level of the system rather than the model, which is the honest claim to make. With the defects from the first round closed, the next steps are the larger pieces the second round began to ask for.

Direction	Planned work
Multi-agent missions	More than one rover in a world, so a class can explore coordination and simple swarm behaviour, which the single rover cannot teach today and which the personas now ask for most
Free spoken dictation	Grow the deterministic voice command set toward free dictation, so a pupil can describe a plan and not only speak the fixed phrases the parser knows
A larger model option	Let a school with a capable machine point the app at a larger local model, lifting the quality ceiling the personas identified without breaking the offline guarantee
Classroom study	An observed study with real pupils under the appropriate ethics approval, which the simulated study can only stand in for

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Generative artificial intelligence assisted the development of the software and the drafting of this report. All technical claims reflect the running system, the evaluation figures are reported as measured from the simulated persona study, and every reference above was checked in a browser at the time of writing.